

CSC1005/CSC1022 Database Lab

Lab 3: SQL Basics

Goal: Learn simple SQL syntax and how to write your basic queries.

This lab is designed to get students familiar with all 3 datasets. Queries are simple but make use of WHERE, LIKE, DISTINCT and ORDER BY.

You have finished the lab when you have finished all queries.

Section 1. Basketball (Make default database)

1. List the ID of the players who won an award in 1989. (Challenge : Don't display duplicate ID's)

Solution: Select (Distinct) playerID from awards_players;

2. List the ID of the players who scored more than 400 points in less than 900 minutes played during the season 1998

Solution: Select playerID, year , points, minutes from players_teams where points > 400 and minutes < 900 and year = 1998;

playerID	year	points	minutes
curryde01	1998	423	864
wallajo01	1998	411	812

3. Return the name and the number of wins at home of the teams with 'Chicago' in their name during the season 1994.

(Challenge : Return the name of the teams with 'C' in their name during the season 1994 where they won more games at home than away)

Solution: Select name, year, homeWon from teams where name like '%Chicago%' and year = 1994;

name	year	homeWon
Chicago Bulls	1994	28

Challenge Solution: Select name, year, homeWon, awayWon from teams where name like '%C%' and year = 1994 and homeWon>awayWon;

Section 2: Sakila (Make Sakila the default database)

4. Show all films with rating 'PG' or 'PG-13', ordered by length (shortest first) and then title.

**Solution: SELECT title, rating, length FROM film WHERE rating IN ('PG', 'PG-13')
ORDER BY length, title;**

(other answers also accepted using LIKE, =)

5. List all distinct film ratings in the database, sorted alphabetically.

Solution: SELECT distinct rating FROM film order by rating;

6. List all films whose title starts with 'THE', ordered by title.

Solution: SELECT title FROM film where title like 'The%';

THEORY MERMAID

Section 3. World (Make default database)

7. Get a list of cities in 'USA' where the city name contains the letter sequence 'New', ordered by name. (Hint : use country code in city table)

**Solution: SELECT Name, District, Population FROM city WHERE CountryCode = 'USA' AND
Name LIKE '%New%' ORDER BY Name**

8. Find all cities whose name ends with 'burg', ordered by population descending.

**Solution: SELECT Name, CountryCode, Population FROM city WHERE Name LIKE '%burg'
ORDER BY Population DESC;**

9. Which distinct languages represent the majority language in at least one country? Display ordered by percentage descending. (Hint: Use 'countrylanguage' table)

**Solution: SELECT DISTINCT Language, CountryCode, Percentage FROM countrylanguage
WHERE Percentage > 50
ORDER BY Percentage DESC;**